

NorthPeak Retail Group

Company Brief & Project Overview

Data Platform Engineering Simulation | Senior Data Engineer

Quick Facts

Company	NorthPeak Retail Group Ltd (fictional)
Headquarters	Leeds, UK
Stores	247 supermarkets across England, Scotland and Wales
Employees	38,400 total 620 in technology
Online customers	1.4 million registered
Loyalty members	4.8 million RewardsCo members
Annual revenue	Approx. GBP 4.2 billion (fictional)
Your role	Senior Data Engineer, Data Platform Engineering team
Your manager	Aisha Okafor, Data Engineering Manager
Platform budget	GBP 3.2 million approved by the board, March 2026
Hard deadline	31 December 2026 — CTO performance bonus of GBP 125,000 at stake

The Five Source Systems

Five acquisitions between 2018 and 2023 left five disconnected source systems that have never been integrated. Finance runs monthly reconciliations in Excel. The board receives five different versions of total revenue.

System	Tech	Acquired	Data
NP Core (primary)	PostgreSQL 14	Original	Orders, customers, products, inventory, stores
GroceryDirect	MongoDB 6	2018	E-commerce orders, product catalogue, digital vouchers
SupplyLink	SQL Server 2019	2020	DC movements, supplier deliveries, stock tracking
RewardsCo	REST API	2021	Loyalty transactions, member profiles, redemptions
NP Financial	Azure SQL (FCA)	2023	Payment processing, FCA-regulated card data

Three Board Escalation Events

The following three incidents in 2025 forced the board to approve GBP 3.2M for a data platform.

1. Christmas 2025 Revenue Gap

Finance reported GBP 312M revenue. E-commerce reported GBP 335M. A GBP 23M discrepancy with no explanation. The CFO escalated to the board. No single source of truth existed.

2. Birmingham DC Inventory Write-off

A silent failure in the SupplyLink-to-EPOS pipeline caused GBP 6.8M of inventory to be written off at the Birmingham distribution centre. Stock arrived at the DC but never appeared in the NP Core order system. The failure was invisible until physical stocktake.

3. GDPR DSAR Failure

The ICO contacted NorthPeak about a subject access request that could not be fulfilled within 30 days because loyalty data (RewardsCo) and order data (NP Core) were in disconnected systems with no unified customer identity. Regulatory fine risk: GBP 500K.

The Team

Aisha Okafor	Data Engineering Manager. 9 years experience. Previously built Ocado's first Airflow implementation.
Dmitri Volkov	Senior Data Engineer — leaving in 8 weeks for Tesco Edinburgh. Documented stages 1-3 only.
Yemi Adeyemi	Analytics Engineer. Has been blocked for 3 months waiting for clean Silver data. Needs FactSales.
Chloe Tan	BI Developer. Hard deadline: Christmas trading dashboard live by 15 November 2026. Needs FactSales.
Marcus Webb	Head of Data Engineering. Signs off on platform certification. Not in daily contact but reviews the code.

Known Data Quality Issues

These issues are documented in Dmitri's process notes and in the seed data. Your pipeline must handle all of them explicitly.

Issue	Location	Scale	Required Treatment
Null customer_id	NP Core orders	4.1% of rows	Sentinel -1 (UNKNOWN_CUSTOMER) + boolean flag. Do NOT drop.
Duplicate order_id	NP Core orders	2.3% (28 dupes)	Window dedup: keep latest by created_at DESC.
Card data in payment_ref	NP Core orders	Pre-2021 records	SHA-256 hash in Silver. Never delete — FCA audit trail required.
Null email	NP Core customers	~35% (IDs 401-470)	FreshMart migration batch. Do not impute — data does not exist.
Mixed date formats	SupplyLink	All records	YYYY-MM-DD and DD/MM/YYYY both present. Standardise to ISO 8601 UTC.
Mixed timezones	SupplyLink	All records	UTC (Oct-Mar) and BST/UTC+1 (Apr-Sep). Normalise all to UTC.
Negative inventory	NP Core inventory	~12 records	Valid DC adjustments. Preserve with is_adjustment=True flag.
Silent SKU failure	SupplyLink reached epoch (wks 38-40)	4%	The weeks 38-40 trap. DO NOT conclude demand drop without checking this.

Medallion Architecture — Bronze, Silver, Gold

Layer	Tool	Purpose	Key Rules
Bronze	PySpark + Parquet	Raw immutable ingestion	Append-only per date. Overwrite partition for idempotency. 5 metadata files.
Silver	PySpark + Delta Lake	Cleansed and masked	All 8 DQ fixes applied. SHA-256 PII masking. Dedup. Timezone normalisation.
Gold	dbt + DuckDB	Business aggregates for BI	FactSales, DimCustomer, DimStore, DimProduct, DimDate. Incremental.

Sprint Timeline

Phase	Sprint	Dates	Deliverable
Phase 1	Sprint 0	14-27 Jul 2026	Architecture ADR, environment setup, onboarding
Phase 2	Sprint 1	28 Jul-10 Aug 2026	PostgreSQL source schema + realistic seed data
Phase 3	Sprint 2	11-24 Aug 2026	PySpark Bronze, Silver, Gold medallion pipeline
Phase 4 (GATE)	Sprint 3	25 Aug-7 Sep 2026	Airflow DAG + GitHub Actions CI/CD + pytest tests
Phase 5	Sprint 4	8-21 Sep 2026	dbt transformations + SQL optimisation + lineage
Phase 6	Sprint 5	22 Sep-5 Oct 2026	Great Expectations + data contracts + observability
Phase 7	Sprint 6	6-19 Oct 2026	Engineering review: SQL, architecture, incident diagnosis
Phase 8	Sprint 7	20 Oct-2 Nov 2026	Production runbook + Terraform plan + INC-001 RCA

Business Stakes and SLAs

Board deadline	31 December 2026 — platform must be certified live
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CTO bonus at stake	GBP 125,000 — contingent on platform delivery
Daily data SLA	Gold layer complete before 06:00 UTC — trading standup at 09:00
Bronze SLA	Ingested by 03:30 UTC
Silver SLA	Cleansed by 04:30 UTC
Gold SLA	Marts complete by 06:00 UTC
DSAR compliance	Unified DimCustomer required to fulfil GDPR requests within 30 days
FCA compliance	SHA-256 PII masking on payment_reference before Silver layer
Quarantine SLA	Less than 1% of Bronze rows quarantined per day
Data freshness SLA	Gold freshness under 25 hours at all times

Glossary

Term	Definition
Medallion architecture	Three-layer data design: Bronze (raw), Silver (cleansed), Gold (aggregates)
INC-001	Production incident: 3 days missing FactSales due to BashOperator exit code swallowing
EPOS	Electronic Point of Sale — the store checkout system that records transactions
DC	Distribution Centre — warehouse receiving stock from suppliers
reached_epos	SupplyLink field: Y if stock was confirmed into NP Core EPOS, null if the handoff failed
UNKNOWN_CUSTOMER	Sentinel value -1 used for null customer_id rows — do not drop these rows
DSAR	Data Subject Access Request — GDPR right to access personal data within 30 days
FCA	Financial Conduct Authority — UK regulator requiring card data masking
pipefail	Bash flag set -e -o pipefail — causes shell to exit on any command failure
Backfill	Re-running a pipeline for past dates: airflow dags backfill ... --reset-dagruns
ADR	Architecture Decision Record — written justification for every major design choice
Grain	The unit one row represents in a fact table (e.g. one row per order vs per order item)

Classification: Internal — SortNext Project Lab | Version 1.0 | July 2026 | NorthPeak Retail Group is a fictional company created for simulation purposes only.